

Modularity of the normal-subobject lattice

A full answer to the conjecture in arXiv:2206.11630

Autonomous proof audit — lane 9

17 August 2026

Candidate full result. For every real seminormed space and every $0 < \varepsilon < 1$, the normal-subobject lattice considered by Fritz is modular if and only if the Hausdorff quotient has dimension at most one. The proof works for arbitrary norms and in arbitrary (possibly incomplete and infinite) dimension.

1 Source conjecture and transcription

Fix $0 < \varepsilon < 1$. Corollary 4.9 of the source identifies the normal subobjects of a seminormed space (V, p) with

$$\mathcal{N}_\varepsilon(V, p) = \{q : q \text{ is a seminorm on } V, p \leq q \leq \varepsilon^{-1}p\},$$

ordered by reverse pointwise order. The meet and join are

$$(q_1 \wedge q_2)(x) = \max\{q_1(x), q_2(x)\}, \quad (q_1 \vee q_2)(x) = \inf_{x_1+x_2=x} (q_1(x_1) + q_2(x_2)).$$

Footnote 12 on printed page 23 conjectures:

$$\mathcal{N}_\varepsilon(V, p) \text{ is modular if and only if } \dim(V/V_0) \leq 1, \text{ where } V_0 = \{x : p(x) = 0\}.$$

2 Idea and geometric translation

Every admissible q has the same null space V_0 as p , so the problem passes exactly to the normed quotient $X = V/V_0$. Write $U = \{x : p(x) \leq 1\}$. The unit ball $K_q = \{x : q(x) \leq 1\}$ is a closed, convex, centrally symmetric set satisfying

$$\varepsilon U \subseteq K_q \subseteq U.$$

Conversely every such set is the unit ball of an admissible norm. Under this correspondence the lattice order is ordinary inclusion and

$$K \wedge L = K \cap L, \quad K \vee L = \overline{\text{conv}}(K \cup L).$$

Indeed, the pointwise maximum has intersection as its unit ball. For the infimal-convolution join, every decomposition of cost at most one gives a convex combination of points of the two balls. Conversely, decompositions of cost at most $1 + \delta$ put $x/(1 + \delta)$ in their convex hull; letting $\delta \downarrow 0$ gives the closure. The join is still bounded between p and $\varepsilon^{-1}p$, so its unit ball is closed in the p -topology. Thus the conjecture becomes a local statement about the lattice of symmetric convex bodies in an arbitrarily thin annulus.

which is the restriction of the lattice structure on the set of *all* seminorms on the vector space V . Unfortunately, $\mathfrak{nSub}(V)$ is typically¹² not modular. For example on $\mathbb{R}^2$, consider the following scalar multiples of the standard p -norms,

$$\frac{1}{4}\|\cdot\|_1, \quad \frac{1}{3}\|\cdot\|_\infty, \quad \frac{1}{5}\|\cdot\|_1.$$

Per Dummy 4.9, these are in the normal subobject lattice on $\mathbb{R}^2$ with respect to any sufficiently small reference norm and any sufficiently small ε . Then, considering the polyhedral unit balls and their vertices as in Figure 1, some calculation shows that

$$\begin{aligned} \left(\frac{1}{4}\|\cdot\|_1 \vee \left(\frac{1}{3}\|\cdot\|_\infty \wedge \frac{1}{5}\|\cdot\|_1\right)\right) \left(\frac{7}{2}, \frac{3}{2}\right) &= \frac{17}{16}, \\ \left(\left(\frac{1}{4}\|\cdot\|_1 \vee \frac{1}{3}\|\cdot\|_\infty\right) \wedge \frac{1}{5}\|\cdot\|_1\right) \left(\frac{7}{2}, \frac{3}{2}\right) &= 1. \end{aligned}$$

Modularity fails since these results are not equal.

Our final goal is to sketch some basics of ε -curved homological algebra in our framework. Our ideal of null morphisms $\mathcal{N}_\varepsilon$ has been chosen precisely such that chain complexes in $(\mathbf{Norm}, \mathcal{N}_\varepsilon)$ amount to the following.

¹²Although we only present one single counterexample to modularity in what follows, its structure indicates that this counterexample is rather generic. More precisely, we conjecture that $\mathfrak{nSub}(V, \|\cdot\|)$ is modular if and only if $\dim(V/V_0) \leq 1$.

Figure 1: Source evidence: arXiv:2206.11630, printed page 23 and footnote 12.

The key observation is that every norm ball in dimension at least two has a thin exposed cap containing a short line segment. Three subsets of that segment reproduce the elementary nonmodular configuration

$$\{0\} \subset [0, 2], \quad \{3\} \cap [0, 2] = \emptyset, \quad \text{conv}(\{0, 3\}) \cap [0, 2] = [0, 2].$$

Adding a common inner ball and reflecting through the origin makes all three sets absorbing and centrally symmetric without changing the exposed cap.

3 An exposed-cap lemma

Let (X, p) be a real normed space, let $\varphi \in X^*$ satisfy $\|\varphi\| = 1$, and choose numbers $0 < r < c < 1$. Put $H_c = \{x : \varphi(x) = c\}$. For a nonempty compact convex set $S \subset U \cap H_c$, define

$$K(S) = \overline{\text{conv}}(rU \cup S \cup (-S)).$$

Lemma 1 (cap localization). *For every $z \in K(S)$,*

$$\text{dist}_p(z, S) \leq \frac{2[c - \varphi(z)]}{c - r}.$$

In particular, $K(S) \cap H_c = S$.

Proof. First take z in the nonclosed convex hull. After collecting coefficients inside each convex set, write

$$z = \alpha s - \beta t + \gamma b, \quad s, t \in S, \text{quad } b \in rU, \text{quad } \alpha, \beta, \gamma \geq 0, \text{quad } \alpha + \beta + \gamma = 1.$$

Since $\varphi(s) = \varphi(t) = c$ and $\varphi(b) \leq r$,

$$c - \varphi(z) \geq 2c\beta + (c - r)\gamma \geq (c - r)(\beta + \gamma) = (c - r)(1 - \alpha).$$

Also $S \subset U$ and $b \in rU \subset U$, so

$$p(z - s) \leq (1 - \alpha)p(s) + \beta p(t) + \gamma p(b) \leq 2(1 - \alpha).$$

Combining the inequalities proves the estimate. Approximation by convex combinations and compactness of S extend it to the closed convex hull. If $\varphi(z) = c$, the right side is zero, so $z \in S$; the reverse inclusion is immediate. $\square$

4 The characterization

Theorem 1. *Let (V, p) be a real seminormed space, let $V_0 = \{x : p(x) = 0\}$, and fix $0 < \varepsilon < 1$. Then $\mathcal{N}_\varepsilon(V, p)$ is modular if and only if $\dim(V/V_0) \leq 1$.*

Proof. All admissible seminorms vanish exactly on V_0 , so quotienting by V_0 gives a lattice isomorphism with the corresponding norm lattice on $X = V/V_0$. We may therefore assume that p is a norm.

If $\dim X = 0$, the lattice has one element. If $\dim X = 1$, every norm on X is a scalar multiple of p . Hence the admissible norms form a chain (the scalar interval $[1, \varepsilon^{-1}]$, with reversed order), and every chain is modular.

Assume now that $\dim X \geq 2$. Choose

$$\varepsilon < r < c < 1, \text{quad } x \in X, \text{quad } p(x) = 1, \quad \varphi \in X^*, \text{quad } \|\varphi\| = 1, \text{quad } \varphi(x) = 1.$$

Hahn–Banach provides φ . Choose $y \in \ker \varphi$ with $p(y) = 1$ and $h > 0$ so small that $3h < 1 - c$. Then the four points

$$z_j = cx + jhy \quad (j = 0, 1, 2, 3)$$

belong to $U \cap H_c$. In that hyperplane set

$$S_A = \{z_0\}, \quad S_C = [z_0, z_2], \quad S_D = \{z_3\},$$

and define $A = K(S_A)$, $C = K(S_C)$, and $D = K(S_D)$. These are closed, convex, symmetric bodies with

$$rU \subseteq A, C, D \subseteq U.$$

Their gauges therefore satisfy $p \leq q \leq r^{-1}p < \varepsilon^{-1}p$, so they are elements of the required lattice. Moreover $A \subseteq C$.

We compare the two sides of the modular identity at the exposed hyperplane H_c . First,

$$\text{dist}_p(S_D, S_C) = h.$$

Set $\eta = (c - r)h/5$. If a point z belonged to $D \cap C$ with $\varphi(z) > c - \eta$, the cap lemma applied once to D and once to C would give

$$\text{dist}_p(z, S_D) < 2h/5, \text{quad } \text{dist}_p(z, S_C) < 2h/5,$$

contradicting the distance h between the two cap sets. Thus

$$\sup_{z \in D \cap C} \varphi(z) \leq c - \eta.$$

It follows that adjoining $D \cap C$ to A cannot create a new point on the top exposed hyperplane: using the cap lemma for A , or simply taking convex combinations and passing to limits, gives

$$(A \vee (D \wedge C)) \cap H_c = \{z_0\}.$$

On the other hand,

$$(A \vee D) \cap H_c = [z_0, z_3], \quad C \cap H_c = [z_0, z_2].$$

Therefore

$$((A \vee D) \wedge C) \cap H_c = [z_0, z_2].$$

In particular z_1 belongs to the right side but not the left side. Hence

$$A \vee (D \wedge C) \neq (A \vee D) \wedge C \quad \text{despite } A \subseteq C,$$

so the lattice is not modular. □

5 Verification, scope, and novelty

The proof was checked against the source’s reverse seminorm order and its explicit meet/join formulas. The inner radius r may be chosen strictly between ε and 1, so the construction works even when ε is arbitrarily close to 1; no “sufficiently small” caveat remains. The cap estimate supplies the uniform gap needed when closed convex hulls are used, so neither completeness nor finite dimensionality is hidden in the proof. Quotienting by V_0 is exact because every admissible seminorm has the same null space.

Scope. This resolves only the real-scalar conjecture stated in the source. It does not develop the paper’s broader proposed theory of ε -curved group cohomology. The same geometric idea should adapt to complex spaces after realification, but that variant is not claimed here.

Novelty status. Bounded searches by the exact conjectural phrase, title, normal-subobject lattice terminology, and seminorm-lattice modularity found the source paper and general seminorm-lattice literature, but no later resolution or matching exposed-cap proof. Novelty is therefore plausible, not certified; expert and author review are recommended before dissemination.

References

- [1] T. Fritz, *Non-abelian and ε -curved homological algebra with arrow categories*, J. Pure Appl. Algebra 227 (2023), 107314; arXiv:2206.11630, Corollary 4.9 and footnote 12.